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**NMAM INSTITUTE OF TECHNOLOGY, NITTE**  
**Off-Campus Centre of Nitte (Deemed to be University)**  
**Fifth Semester B.Tech (CBCS) Degree Examinations**

December 2025

**EC2221-1 – COMPUTER OPERATING SYSTEMS**

(For EC)

Duration: 3 Hours

Max. Marks: 100

**Note:**

**Part – A: Multiple Choice Questions:** Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

**Part – B: Descriptive Answer Questions:** Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

**PART - A: MULTIPLE CHOICE QUESTIONS****20 Marks**

1. Which of the following is the lowest layer in a layered operating system?  
A) User interface B) Hardware  
C) Application D) Security module
2. Which of the following components is not a part of a monolithic operating system?  
A) File system B) Device drivers  
C) Microservices D) Process management
3. Time-sharing OS is also known as:  
A) Batch processing OS B) Interactive OS  
C) Single-user OS D) Real-time OS
4. Which layer of the operating system interacts directly with hardware?  
A) Application Layer B) User Interface  
C) Kernel D) Middleware
5. System programs in an OS are responsible for:  
A) User interaction B) System booting  
C) System maintenance D) Security only
6. The major drawback of a monolithic kernel is:  
A) Complexity in debugging B) Low performance  
C) Lack of modularity D) Slow execution
7. A program becomes a process when:  
A) It is compiled B) It is stored in memory  
C) It is loaded into memory and starts execution D) It is linked with libraries
8. The PCB does not contain:  
A) Process ID B) Program counter  
C) CPU registers D) Network settings
9. Inter-process Communication mechanisms include:  
A) Shared memory B) File transfer  
C) System reboot D) Process termination
10. What is the primary purpose of the Memory Management Unit in a computer system?  
A) Manage process execution B) Translate logical addresses to physical addresses  
C) Increase CPU speed D) Schedule tasks
11. Which scheduling algorithm is commonly used in UNIX systems?  
A) FCFS B) Round Robin  
C) Priority Scheduling D) Multilevel Queue Scheduling
12. In UNIX, zombie process occurs when:  
A) A process is killed abruptly B) A process completes execution but its PCB remains  
C) A process is in a deadlock D) A process is blocked forever



13. The main disadvantage of contiguous memory allocation is:  
 A) Internal fragmentation B) Slower execution  
 C) Poor security D) High CPU usage
14. Segmentation divides memory into:  
 A) Equal-sized blocks B) Variable-sized segments  
 C) Pages D) Cache
15. What is thrashing in virtual memory?  
 A) Continuous swapping of pages causing high CPU usage B) Efficient execution of processes  
 C) Storing frequently used pages in the cache D) Optimizing memory allocation
16. Which page replacement policy may lead to Belady's anomaly?  
 A) FIFO B) LRU  
 C) Optimal Page Replacement D) Clock algorithm
17. What is the purpose of a page table?  
 A) It maps virtual addresses to physical addresses B) It stores all process control information  
 C) It organizes files on disk D) It manages process scheduling
18. Which of the following is NOT a page replacement policy?  
 A) NRU (Not Recently Used) B) LFU (Least Frequently Used)  
 C) CFS (Completely Fair Scheduler) D) MFU (Most Frequently Used)
19. What is the main disadvantage of demand paging?  
 A) Slower program execution due to page faults B) Increased CPU usage  
 C) Increased disk fragmentation D) High memory wastage
20. Which scheduling algorithm prevents starvation?  
 A) Round Robin B) FCFS  
 C) SRTF D) HRN

### PART - B: DESCRIPTIVE ANSWER QUESTIONS

#### Unit – I

|                                                                                             | Marks | BT* | CO* | PO* |
|---------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1. a) Define operating system. Explain the key operational concerns of an operating system. | 8     | L*2 | 1   | 1   |
| b) Explain the terms: i) Efficiency ii) System Performance iii) User service.               | 8     | L2  | 1   | 1   |
| 2. a) Explain the distributed operating systems in detail. And also mention its benefits.   | 8     | L2  | 1   | 1   |
| b) Explain the structure of kernel-based operating systems.                                 | 8     | L2  | 2   | 1   |
| 3. a) Explain the multiprogramming operating system.                                        | 8     | L2  | 1   | 1   |
| b) With a neat diagram explain the turnaround time in the batch processing system.          | 8     | L2  | 2   | 1   |

#### Unit – II

|                                                                                                                                                           |   |    |   |   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|---|
| 4. a) Explain the different fields of the Process Control Block and their role in process management within an operating system.                          | 8 | L2 | 3 | 1 |
| b) Explain kernel-level threads and discuss their advantages and disadvantages.                                                                           | 8 | L2 | 3 | 1 |
| 5. a) Illustrate the relationship between threads and processes, explaining different implementation models and their arrangement in an operating system. | 8 | L2 | 3 | 1 |
| b) Explain the memory allocation model for a process in an operating system.                                                                              | 8 | L2 | 4 | 1 |
| 6. a) Explain child processes, their creation, and the benefits of using them in an operating system.                                                     | 8 | L2 | 3 | 1 |
| b) Explain the physical organization of memory in paging and describe how a logical address is translated into a physical address.                        | 8 | L2 | 4 | 1 |

## Unit – III

7. a) Describe various page replacement policies in detail, explaining their advantages, and disadvantages.
- b) For the following set of Process perform FCFS and SRN scheduling to calculate mean turnaround time and mean weighted turnaround.

8 L2 5 1

| Process        | P1 | P2 | P3 | P4 | P5 |
|----------------|----|----|----|----|----|
| Admission Time | 0  | 2  | 3  | 5  | 9  |
| Service Time   | 3  | 3  | 2  | 5  | 3  |

8 L3 5 2

8. a) Explain the concept of demand paging and describe the steps involved in handling a page fault with the help of an appropriate diagram.
- b) For the following set of process perform Least Completed Next (LCN) scheduling to calculate mean turnaround time and mean weighted turnaround. Also, construct the Gantt Chart to visualize the scheduling sequence.

8 L2 5 1

| Process        | P1 | P2 | P3 | P4 | P5 |
|----------------|----|----|----|----|----|
| Admission Time | 0  | 2  | 3  | 4  | 8  |
| Service Time   | 3  | 3  | 5  | 2  | 3  |

8 L3 5 2

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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